FLUOR DANIEL

BOLTED END PLATE MOMENT CONNECTIONS

PURPOSE

This practice establishes guidelines for the design of bolted end plate moment connections.

SCOPE

This practice includes information about the design of the following 3 types of end plate beam-to-column flange moment connections:

- Standard 4 tension bolt connection (Refer to Attachment 01, Figure 1.)
- Haunched 4 tension bolt connection (Refer to Attachment 01, Figure 2.)
- Stiffened 8 tension bolt connection (Refer to Attachment 01, Figure 3.)

Procedures and sample calculations are based on AISC (American Institute for Steel Construction), Manual of Steel Construction, ASD 9'th Edition., working stress design.

APPLICATION

Moment connections may be used on piperacks, equipment structures, and buildings.

Standard 4 Tension Bolt

Standard 4 tension bolt connections (total number of bolts is 6 or 8) are the most common connection for the type of structures designed at Fluor Daniel. They are easy to fabricate and erect, and are the most economical type of moment connection. For large moments, use of a deeper beam or deviation from standard AISC horizontal gage will help in the design of standard 4 tension bolt connection. Other types of connections may be used if this type is inadequate.

Haunched 4 Tension Bolt

Haunched 4 tension bolt connections may be used if column stiffeners interfere with out-of-plane beams framing into column at same elevation. Deepening the connection with a haunch will reduce beam flange forces and may eliminate the need for column stiffeners. This connection is also used to reduce excessive column flange bending and column web shear. Haunched connections are unsightly and more costly to fabricate. They should be avoided if possible.

Stiffened 8 Tension Bolt

Stiffened 8 tension bolt connections (total number of bolts is 12 or 16) will carry larger moments than the standard 4 tension bolt connection. The tension load is spread out over a wider area of the column flange and may eliminate the need for stiffeners in the column. However, this connection requires a stiffener at the end plate that may reduce available space above the connection for piping, access, and other similar functions. At present, only A 325 bolts and A 36 steel may be used for this type of connection.

BOLTED END PLATE MOMENT CONNECTIONS

Bolt forces, column web shear, and column flange bending may be excessive for very large moments. For this, a haunched, stiffened, 8 tension bolt connection may be used. The bottom bolts could be 4 tension if adequate.

LOADING

General

Moment connections will be designed for gravity and lateral (wind or seismic) loads. Lateral loads may cause moment reversals (that is, tension at the bottom beam flange).

Static Loading

The design procedure presented in this practice is for static loading only. The procedure is applicable to repeated loads such as live and wind, usually considered as static equivalents in conventional analysis. The procedure is not applicable to dynamic loading. Test data are limited for performance of this connection under repeated and cyclic load.

Seismic

End plate type connections can be used in seismic areas. Design procedure in this practice is applicable to seismic zones 0, 1, and nonductile moment resisting frames in Zone 2. For ductile moment resisting frames in Zone 2 and moment resisting frames in Zones 3 and 4, special ductility requirements in the latest edition of the UBC (Uniform Building Code) should be reviewed.

Note!!! Welded type moment connections are normally used in structures over 50 feet high in Zones 3 and 4.

COMPUTER DESIGN

Design of end plate connections should be done using electronic spreadsheets or similar computer programs if available. Consult the Lead Structural Engineer for availability of computer programs.

CONNECTION DETAILS

Design and detailing will meet the requirements of AISC specifications.

A325-N 3/4 inch diameter bolts will normally be used. Larger diameter bolts, up to 1-1/2 inch, and A490 bolts may be used for heavily loaded connections. Only A325 bolts may be used for stiffened 8 tension bolt connection.

The usual vertical bolt gage (g_v) is 3- 1/2 inches for beam flange thickness up through 1/2 of an inch. For flange thickness greater than 1/2 of an inch, the gage is 4 inches. For stiffened 8 tension bolt connection, refer to Attachment 01, Figure 3.

The horizontal bolt gage (g_h) should be standard AISC gage. Where excessive column flange bending occurs, horizontal gage may be reduced. The minimum gage will be 3- 1/2 inches for column web thickness up through 1/2 of an inch, and 4 inches for web thickness greater than 1/2 of an inch. For stiffened 8 tension bolt connection, horizontal bolt gage must be between 5- 1/2 and 7- 1/2 inches.

Width of end plate should be a minimum of 1 inch larger than beam flange width, and preferably should not exceed column flange width. End plate width should be increased if required for bolt edge distance.

Minimum end plate thickness will be the same as the bolt diameter. Thin end plates result in loss of connection rigidity and may induce bolt prying forces.

Column stiffener plates, if required by calculations, will be the same thickness as beam flange rounded down in 1/8 of an inch increments. Stiffeners should preferably be full column depth.

Column webs may be reinforced with doubler plates or diagonal stiffeners if required. Doubler plates increase fabrication costs, require much welding, and may cause distortion of column flanges. Diagonal stiffeners also increase fabrication costs and may interfere with other framing connections. Generally, it is less costly to select column members with adequate web thickness. Special detailing requirements for web reinforcing are found in Blodgett, Design of Welded Structures and AISC, Engineering For Steel Construction.

Column flanges may be reinforced if excessive flange bending exists. Details of column flange reinforcement and capacity are found in Mann and Morris, Limit Design of Extended End Plate Connections.

DESIGN PROCEDURE

Notation For Standard Connections

α_m Coefficient applied to determine bending moment

A_f Area of tension flange, beam
A_{st} Area of pair of column web stiffeners
A_w Area of web clear of flanges, beam

b_{ep} End plate width
b_f Effective column flange bending length
b_{fb} Beam flange width

BOLTED END PLATE MOMENT CONNECTIONS

C_a	1.13 for F_y = 36ksi or 1.11 for F_y = 50ksi
C_b	$\sqrt{\frac{b_{fb}}{b_{ep}}}$
c_{fw}	Column flange width
D	Depth for haunch connection
D_w	Weld size in sixteenths
d'	Bolt hole diameter, d_{bo} + 1/16 inches
d_b	Beam depth
d_{bo}	Bolt diameter
d_c	Column depth
e	Eccentricity of axial load in haunch connection
F	Flange force computed from moments
F_c	Flange force, compression, computed from moments with axial load
f_t	Bolt tensile stress
F_t	Flange force, tension, computed from moments with axial load
f_v	Bolt shear stress
F_v	Allowable sheer stress 0.4 F_y
F_y	Steel yield stress
g_h	Horizontal bolt gage
g_v	Vertical bolt gage
k	Vertical toe fillet for column, AISC W shapes dimensions
k_1	Horizontal toe fillet for column, AISC W shapes dimensions
M	Beam moments for design of connection
M′	Modified moment for haunched connection to account for axial load eccentricity
M_e	Bending moments to determine end plate thickness
M_{eh}	Column flange bending moment
P	Beam axial load for design of connection
P_b	Distance between top 2 rows of bolts, 8 tension bolt connection
P_{bf}	Factored flange force 4/3 F_c or 4/3 F_t for dead, live with wind or earthquake 4/3 F_c or 4/3 F_t for dead and live only For seismic zones 2, 3, and 4, refer to UBC (Uniform Building Code)
P_e	Distance for computation of moments for end plate thickness
P_{eh}	Distance for computation of column flange bending moments
P_f	Distance from top of tension flange to bolt centerline
P_T	Bolt pretension AISC Table J3.7, 8 tension bolt connection

BOLTED END PLATE MOMENT CONNECTIONS

T	Column web depth clear of fillets AISC W shapes dimensions
t_{ep}	End plate thickness
t_{fb}	Beam flange thickness
t_{fc}	Column flange thickness
t_{fh}	Haunch flange thickness
t_{p1}	End plate thickness for stiffness, 8 tension bolt connection
t_{p2}	End plate thickness for strength, 8 tension bolt connection
t_s	End plate/beam flange stiffener thickness, 8 tension bolt connection
t_{st}	Thickness of column web stiffeners
T_u	Ultimate bolt force, 8 tension bolt connection
t_{wb}	Beam web thickness
t_{wc}	Column web thickness
t_{wh}	Haunch web thickness
V	Beam shear for design of connection
V_s	Column story shear, AISC commentary E6
w	Weld size, beam flange to end plate

Note!!! This is the notation used in computation of stiffened flange column capacity for the standard or haunched 4 tension bolt connection.

ASCE ST 3: Limit design of extended end plate connection by A. P. Mann and L. J. Morris, March, 1979.

A	Horizontal bolt gage, g_h
B	End plate width, b_{ep}
C	Vertical bolt gage, g_v
D	Bolt diameter, d_{bo}
D′	Bolt hole diameter d′
F_{ms}	Yield capacity of stiffened column flange
m	$g_h/2$ – column k_1
n	Distance from bolt centerline to edge of end plate
n′	Distance from bolt centerline to edge of column flange
v	Distance from top of tension flange to bolt centerline, P_f
w	$[m (m + n' - 0.5D')]^{1/2}$

Standard 4 Tension Bolt

Design procedure will be according to AISC (9th Edition, pp. 4-116 to 4-122), with the appropriate table numbers, equation numbers, and paragraph numbers, indicated parenthetically below:

1. Calculate beam flange force from computed moments and axial force.

$$\text{Flange force, } F = \frac{M}{(d_b - t_{fb})} \pm \frac{P}{2}$$

BOLTED END PLATE MOMENT CONNECTIONS

2. Calculate bolt tension force or stress assuming the 4 bolts carry the tension equally.

   Calculate bolt shear force or stress assuming all (6 or 8) bolts carry the shear equally.

   Verify that bolt forces do not exceed allowables of AISC (Table J 3.2 and J 3.3 for combined shear and tension).

3. Maximum effective end plate width is beam flange width plus 1 inch. End plate width should preferably not exceed column flange width. Check bolt edge distance according to AISC (Table J 3.5).

   Calculate end plate thickness according to AISC procedure (p. 4-119). Minimum thickness of end plate will be same as bolt diameter.

4. Calculate required beam top flange to end plate fillet weld using actual flange force. Normal welding electrodes are E70xx. Weld size must meet AISC minimum size (Table J 2.4).

5. Calculate beam web to end plate weld. This should develop maximum web tension (0.6 Fy) in the web near the flanges. The minimum size of fillet weld will meet AISC requirements.

6. Investigate column web shear following AISC (commentary E6).

$$\frac{F}{d_c t_{wc}} = \frac{\frac{M}{d_b - t_{fb}} \pm \frac{P}{2}}{d_c t_{wc}} \le F_v = \frac{\text{Allowable Shear Stress}}{\text{(AISC, equation F4-1)}} = 0.4\ F_y$$

   Also consider story shear (V_s) from above or below the joint. The story shear may increase or decrease flange force, F.

7. Investigate requirement for column stiffeners opposite the compression flange for column web yielding (AISC p. 4-117).

$$P_{bf} \le F_{yc} t_{wc} (t_{fb} + 6k + 2t_{ep} + 2w)$$

   where

   P_{bf} = 4/3 x computed flange force (dead, live, and wind or earthquake)
   = 5/3 x computed flange force (dead and live only)

   For seismic Zones 2, 3, and 4, refer to the latest UBC code for ductility requirements.

8. Investigate requirement for column stiffeners opposite the compression flange for column web buckling (AISC 9th Edition, Equation K1-8).

$$P_{bf} \leq \frac{4100\, t_{wc}^3 \sqrt{F_y}}{T}$$

where

T = column web depth clear of fillets as in AISC W shapes.

9. Investigate requirement for column stiffeners opposite the tension flange for overall column flange bending according to AISC procedure (pp. 4-119 to 4-122).

10. For overall column flange bending, the AISC procedure specifies only whether stiffeners are required, but gives no means of determining the capacity of the column flange with stiffeners added. Column flange capacity with stiffeners is computed as follows:

$$F_{ms} = \left[\left(\frac{1}{V} + \frac{1}{w}\right)(2m + 2n' - D') + (2v + 2w - D')\left(\frac{1}{m}\right)\right] t_{fc}^2 F$$

$$w = [m(m + n' - 0.5D')]^{\frac{1}{2}}$$

Allowable capacity for stiffened flange is 0.75 F_{ms}. For notation, refer to Attachment 1, Figure 5.

In addition to the above, the minimum column flange thickness if stiffeners are required will be according to the ECCS (European Code) equation modified for working stress design.

$$\text{Minimum } t_{fc} = 0.3\sqrt{\frac{P_{bf}}{F_y}} \text{ inches}$$

This formula is similar to AISC formula K1-1 (9th Edition) for unstiffened flanges.

11. Prying forces from AISC (pp. 4-90 to 4-95).

End plates sized according to AISC procedure should result in negligible prying forces at working stress. Thick end plates reduce bolt prying action and are ideal for connection stiffness. For end plates with thicknesses greater than or equal to the bolt diameter, prying forces need not be checked.

Haunched 4 Tension Bolt

Behavior of the haunched connection is similar to the standard 4 tension bolt connection. Haunched connections increase the effective depth of the beam, and thus reduce the flange and bolt forces.

Haunched connections reduce column flange bending and web shear, and may also eliminate the requirement for column stiffeners. However, the fabrication costs are substantial. Haunched connections are unsightly and their use should be restricted. Depth of haunched connection (D) will normally be 1- 1/2 the nominal depth of the beam.

Length of haunch is same as nominal depth of beam.

Width and thickness of the haunch flange is same as the beam flange with the thickness rounded down to the nearest 1/8 of an inch.

Haunch web thickness is the same as beam web.

Total number of bolts is 8.

Design procedure is same as standard 4 tension bolt connection with exception of the following:

- Nominal depth of beam is ($D - t_{fb}$)
- For significant axial load, the applied moments are modified to account for the eccentricity introduced by the haunch.

$$\text{Modified Moment } M' = M + Pe$$

$$\text{Where } e = \frac{(D - d_b)}{2}$$

For Fluor Daniel Standard Haunch:

$$D = 1.5\, d_b$$

$$e = 0.25\, d_b$$

Stiffened 8 Tension Bolt

Use of this connection is limited to A 36 steel and A 325 bolts. Total number of bolts for this connection is 12 or 16 depending on magnitude of moment reversals. (Wind moments combined with dead load moment may result in larger tension at top flange. The top flange in tension may require 8 bolts, the bottom flange in tension may require only 4 bolts.)

This connection requires use of stiffener between beam flange and end plate. Thickness of this stiffener should be equal to or greater than thickness of beam web rounded to 1/8 of an inch.

Design procedure presented here is based on Murray and Kukreti, Design of 8-Bolt Stiffened Moment End Plates. An alternative short but conservative procedure is in the AISC Manual (pp. 4-122 to 4-125). Detailed procedure can be easily performed with computer spreadsheets:

1. Calculate beam flange force.

$$\text{Flange Force, } F = \frac{M}{d_b - t_{fb}} \pm \frac{P}{2}$$

2. Estimate bolt tension force as flange force $\frac{F}{6.8}$.

   Calculate bolt shear assuming all 12 or 16 bolts carry the shear equally.

   Select trial bolt size. As a first trial, bolt size should be 3/4 of an inch diameter, and combined shear and tension should not exceed AISC allowables (Tables J 3.2 and J 3.3).

3. Calculate end plate thickness from stiffness criterion to ensure a rigid connection.

$$t_{p1} = \frac{0.00885\, P_f^{0.873}\, g_h^{0.577}\, F^{0.917}}{d_{bo}^{0.924}\, t_s^{0.112}\, b_{ep}^{0.682}}$$

4. Calculate end plate thickness from the strength criterion.

$$t_{p2} = \frac{0.00625\, P_f^{0.257}\, g_h^{0.148}\, F^{1.017}}{d_{bo}^{0.719}\, t_s^{0.162}\, b_{ep}^{0.319}}$$

5. End plate thickness is greater of t_{p1} and t_{p2} rounded to nearest 1/8 of an inch. Minimum thickness will be same as bolt diameter.

6. Calculate ultimate bolt force. Maximum bolt force includes prying action effects and bolt pretension, P_T (Table J 3.7).

$$\text{Ultimate Bolt Force, } T_u = \frac{1.381 \times 10^{-4}\, P_f^{0.591}\, F^{2.583}}{t_{ep}^{0.885}\, d_{bo}^{1.909}\, t_s^{0.327}\, b_{ep}^{0.965}} + P_T$$

BOLTED END PLATE MOMENT CONNECTIONS

7. Check bolt size.

$$T_u < 2 * (T_{allowable})$$

$T_{allowable}$ = Allowable tension load from AISC, including combined shear and tension

If the above requirement cannot be obtained, then increase end plate thickness to reduce T_u or select larger bolt diameter.

8. Beam top flange to end plate weld and beam web to end plate weld are same as that of standard 4 tension bolt connection.

9. Stiffener to end plate and beam flange weld is same as the beam web to end plate web.

10. Investigate column web shear. This is the same as standard 4 tension bolt connection.

11. The requirement for column stiffeners opposite compression flange for column web yielding and buckling is the same as standard 4 tension bolt connection.

12. Investigate the requirement for column stiffeners opposite tension flange for overall column flange bending according to AISC procedure (pp. 4-125). Horizontal gage is between 5- 1/2 and 7- 1/2 inches. Local bending Equation K1-1 does not apply to this connection. The flange force is spread out over a larger area.

    Calculation of capacity of stiffened flange is currently unavailable. Column flange bending is less severe for this connection compared to standard 4 tension bolt connection. Tension force is spread out over a larger area.

REFERENCES

AISC (American Institute for Steel Construction), Chicago: Allowable Stress Design, 9th Edition; 1989: 6-14 - 6-27.

ECCS (European Code)

UBC (Uniform Building Code, 1994 Edition)

Blodgett, Omer M. Design of Welded Structures. Cleveland, OH: The James F. Lincoln Arc Welding Foundation. 1966.

Curtis, Larry E., and Thomas M. Murray. Column Flange Strength at Moment End Plate Connections. AISC Engineering Journal 2. Pages 41-49. 1989.

Kulak, G. L., J. W. Fisher, and J. H. Struik. Guide to Design Criteria for Bolted and Riveted Joints. 3rd. Edition, Chapters 16 to 18. New York: John Wiley and Sons. 1987.

Mann, Alan P., and Linden J. Morris. Limit Design of Extended End Plate Connections. ASCE Journal. Pages 511-527. March, 1989.

Murray, Thomas J. Moment Connection. Journal of Constructional Steel Research, Vol. 10. Pages 142-161. 1988.

Murray, Thomas M., and Anant R. Kukreti. Design of 8-Bolt Stiffened Moment End Plates. AISC Engineering Journal. Pages 45-52. 1988.

Tsai, K. C., and E. P. Popov. Cyclic Behavior of End Plate Moment Connections. ASCE Journal, Pages 2917-2931. November, 1990.

Witteveen, Jelle, et al. Welded and Bolted Beam to Column Connections. ASCE Journal. Pages 433-455. February, 1982.

ATTACHMENTS

Attachment 01:
Moment Connections Figures 1 Through 5

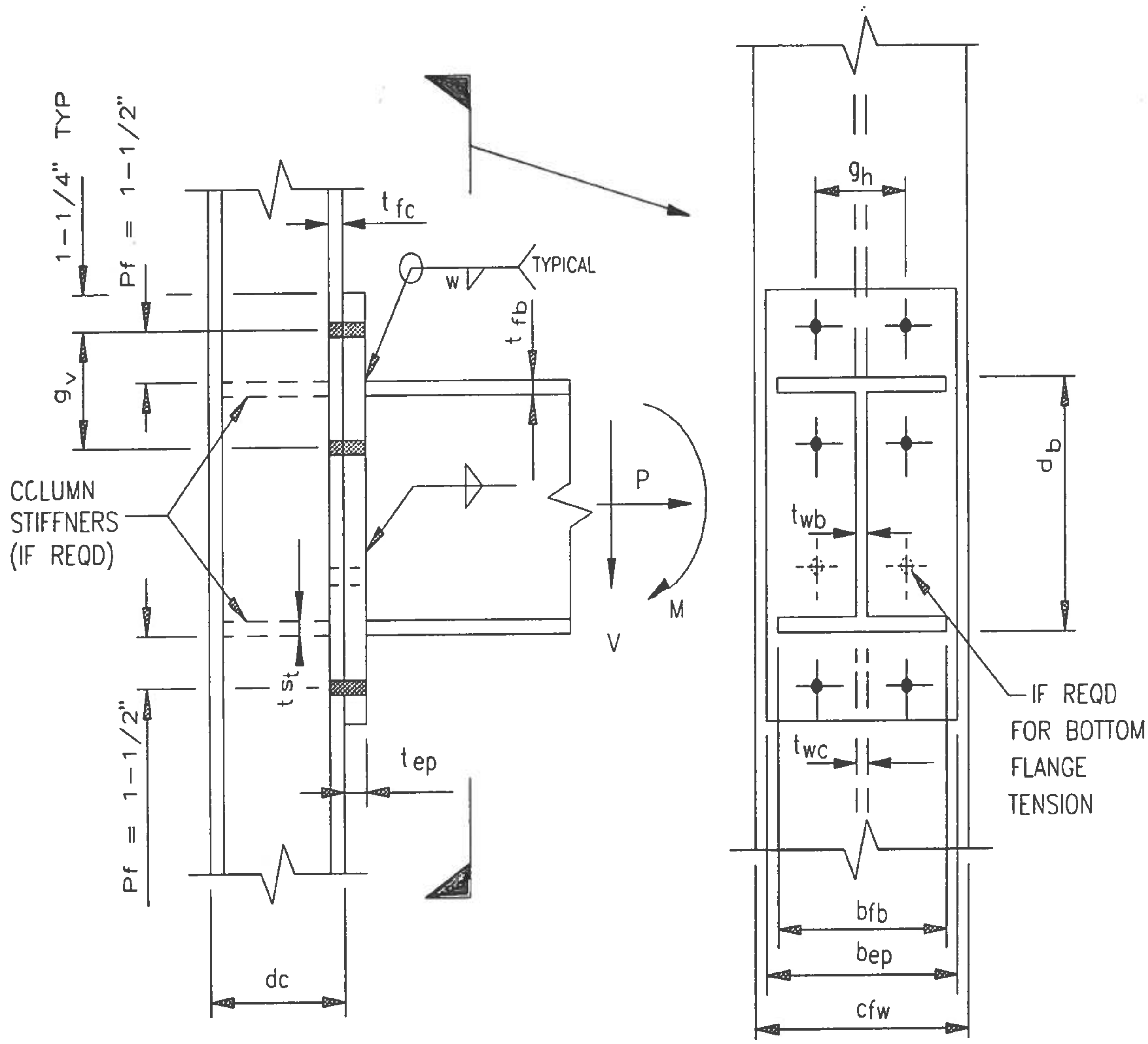
Attachment 02:
Sample Design 1: End Plate Moment Connection, Standard Four Tension Bolt

Attachment 03:
Sample Design 2: End Plate Moment Connection, Haunched Four Tension Bolt

Attachment 04:
Sample Design 3: End Plate Moment Connection, Stiffened Eight Tension Bolt

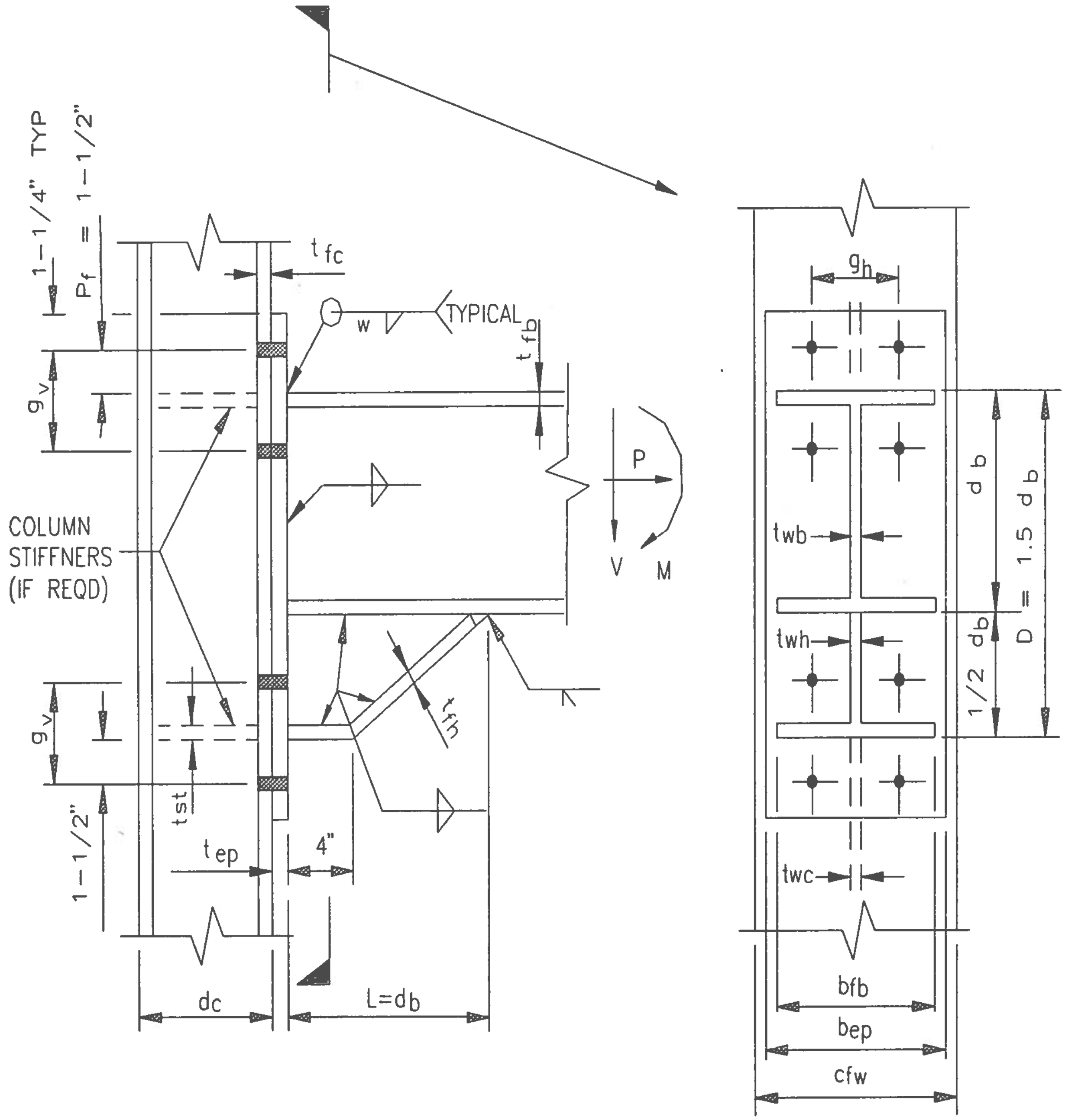
MOMENT CONNECTIONS, FIGURES 1 THROUGH 5

Figure 1: Standard 4 Tension Bolt Connection



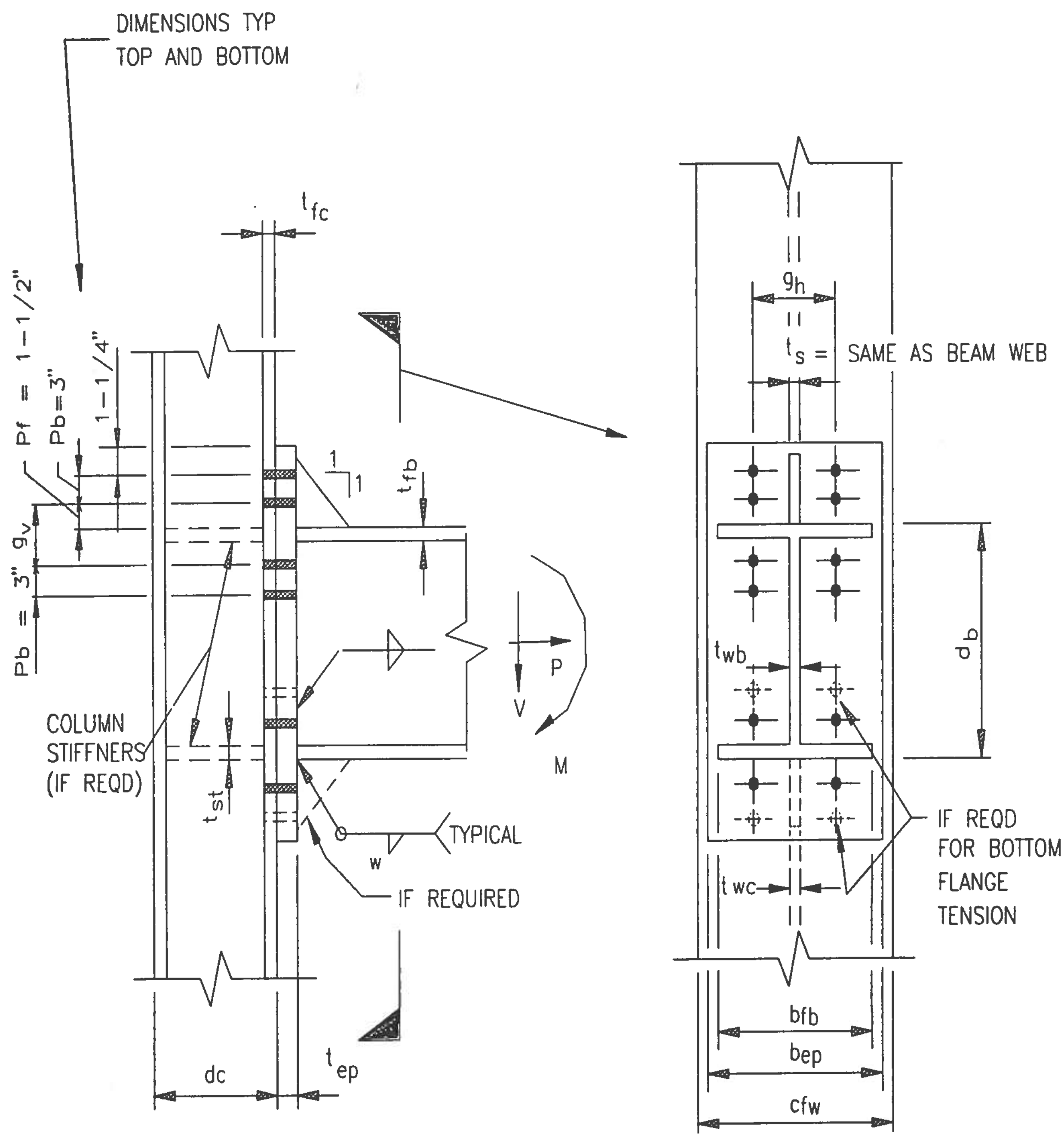
MOMENT CONNECTIONS, FIGURES 1 THROUGH 5

Figure 2: Haunched 4 Tension Bolt Connection



MOMENT CONNECTIONS, FIGURES 1 THROUGH 5

Figure 3: Standard 8 Tension Bolt Connection



MOMENT CONNECTIONS, FIGURES 1 THROUGH 5

Figure 4:

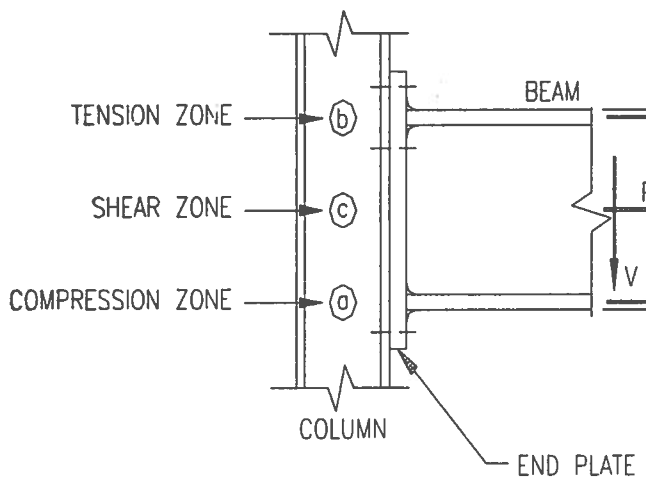
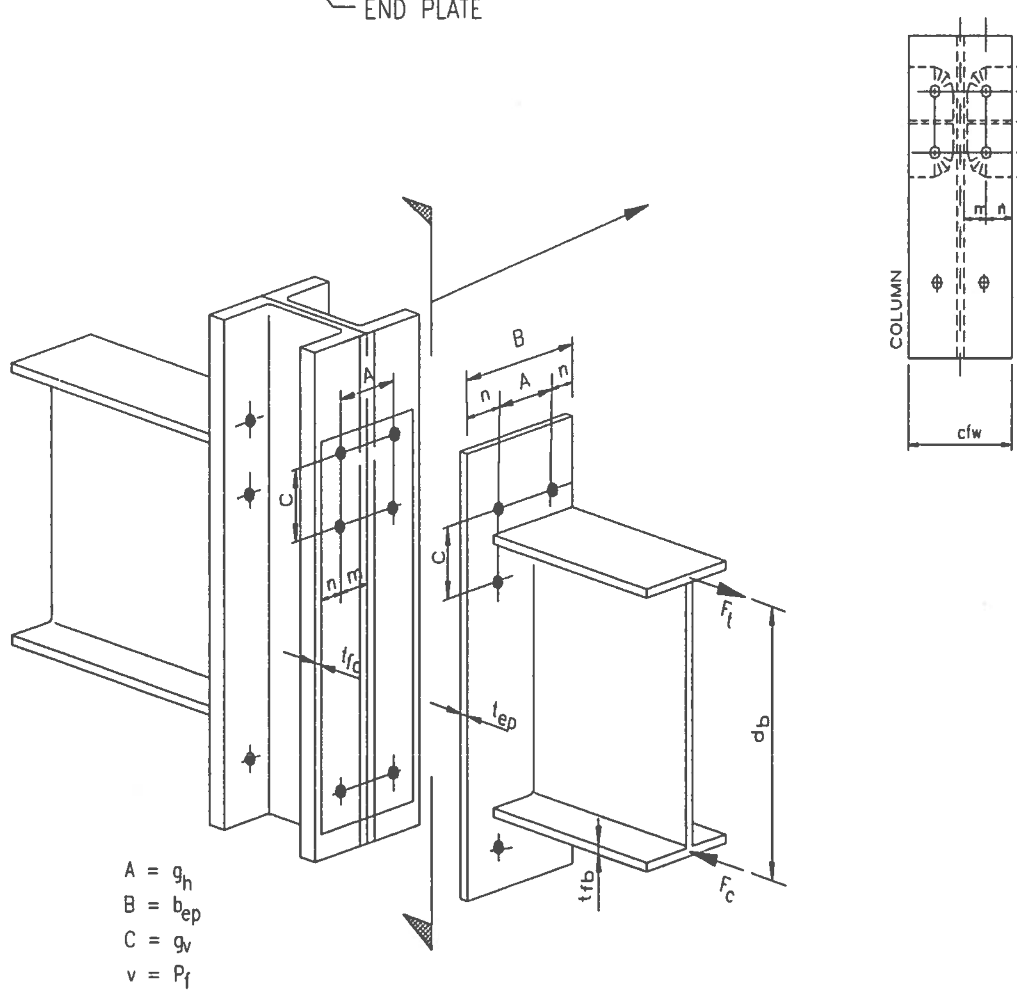


Figure 5:



SAMPLE DESIGN 1 : END PLATE MOMENT CONNECTION--STANDARD FOUR TENSION BOLT

DESIGN DATA (Refer to Figure 1)

BEAM SIZE: W12 X 26	COLUMN SIZE: W10 X 33
$d_b = 12.22$ in	$d_c = 9.73$ in
$b_{fb} = 6.49$ in	$c_{fw} = 7.96$ in
$t_{fb} = 0.38$ in	$t_{fc} = 0.435$ in
$t_{wb} = 0.23$ in	$t_{wc} = 0.290$ in
	$k = 1\frac{1}{16} = 1.063$ in
	$k_1 = \frac{11}{16} = 0.688$ in

$g_v = 1.5" + 2" = 3.5"$

$g_h = 5.5"$ (Standard gage for W10 X 33 column)

Bolts: A325N $\frac{3}{4}"\varnothing$ Total # of bolts = 6 $F_Y = 36$ KSI

LOADS:

Beam Shear (V) = 12.8K
Beam Moments (M) = $45.0^{'-K}$
Beam Axial (P) = 2.0K (Tension)

NOTE: The above loads are for operating plus wind and have not been reduced.

The connection design forces are reduced by 75%. This is equivalent to $\frac{1}{3}$ increase in allowable stresses.

Moments,	$M = 0.75 \times 45.0 = 33.75^{'-K}$
Axial,	$P = 0.75 \times 2.0 = 1.5K$ (Tension)
Shear,	$V = 0.75 \times 12.8 = 9.6K$

NOTE: Refer to the standard connection detail and call out on engineering sketches

1) **BEAM FLANGE FORCE, F**

$$F = \frac{M}{(d_b - t_{fb})} \pm \frac{P}{2}$$

Tension Flange, $F_t = \dfrac{(33.75)(12)}{12.22 - 0.38} + \dfrac{1.5}{2} = 34.20 + 0.75 = 35.0K$

Compression Flange, $F_c = 34.20 - 0.75 = 33.46K$

2) **BOLT STRESSES**

Bolt Shear Stress $= \dfrac{9.6}{(6)(0.4418)} = 3.6$ KSI < 21.0 KSI (Table J3.2 Threads not excluded)

Allowable Tensile Stress $= \sqrt{44^2 - 4.39(f_v)^2} = \left(\sqrt{[44(4/3)]^2 - 4.39(\frac{3.6}{0.75})^2}\right)(0.75) = 43.35$ KSI

SAMPLE DESIGN 1 : END PLATE MOMENT CONNECTION–STANDARD FOUR TENSION BOLT

(Table J3.3)

Bolt Tensile Stress = $\frac{35.0}{(4)(0.4418)}$ = 19.8 KSI < 43.35 KSI

3) END PLATE

End Plate width (maximum effective), $b_{ep} = b_{fb} + 1" = 6.49" + 1.0" = 7.49"$

Use 8" end plate width for bolt edge distance of $1\frac{1}{4}$" (Table J3.5) and $5\frac{1}{2}$" gage. Maximum effective width for stress is 7.49".

End plate thickness (AISC PP 4-119 and A-120 for procedure)

$P_e = P_f - (d_{bo} / 4) - .707w = 1.5 - (\frac{3}{4})/4 - 0.707(\frac{1}{4}") = 1.14"$

$C_a = 1.13$; $C_b = \sqrt{b_{fb} / b_{ep}} = \sqrt{6.49 / 7.49} = 0.93$

A_f = Area of tension flange = $b_{fb}\, t_{fb} = (6.49)(0.38) = 2.47\ in^2$

A_w = Web Area clear of flanges = $(d_b - 2t_{fb})(t_{wb}) = (12.22 - 2(0.38))(0.23) = 2.64$

$\alpha_m = C_a C_b (A_f / A_w)^{1/3} (P_e/d_{bo})^{1/4} = (1.13)(0.93)(2.47 / 2.64)^{1/3} (1.14 / 0.75)^{1/4} = 1.14$

$M_e = \alpha_m F_t P_e / 4 = (1.14)(35.0)(1.14) / 4 = 11.4^{"\text{-}K}$

End Plate Thickness Required $= \sqrt{\frac{6M_e}{(WIDTH)(0.75F_Y)}} = \sqrt{\frac{6(11.4)}{(7.5)(27)}} = 0.58"$

$\boxed{\text{USE MINIMUM 3/4"}}$

4) END PLATE TO BEAM FLANGE FILLET WELD

This is sized using tension flange force.

$D_{W(REQD)} = \frac{F_t}{[2(b_{fb} + t_{fb}) - t_{wb}]0.928} = \frac{35.0}{[2(6.49 + 0.38) - 0.23]0.928}$ = 2.79 Sixteenths

Minimum Weld Size (Table J2.4) = $\frac{1}{4}$" (Note: End Plate Thickness is 3/4")

5) END PLATE TO BEAM WEB FILLET WELD

This should develop maximum web bending stress near flange

$D_{W(REQD)} = \frac{0.6F_y t_{wb}}{(2)(0.928)} = \frac{0.6(36)(0.23)}{(2)(0.928)}$ = 2.68 Sixteenth

Minimum Weld Size = $\frac{1}{4}$"

SAMPLE DESIGN 1 : END PLATE MOMENT CONNECTION–STANDARD FOUR TENSION BOLT

6) COLUMN WEB SHEAR (Commentary E6)

Maximum Flange Force = 35.0K

$$\text{Column Web Shear} = \frac{F}{d_c t_{wc}} = \frac{35.0}{(9.73)(0.29)} = 12.40 \text{ KSI} < 0.4(36) = 14.4 \text{ KSI}$$

7) COLUMN STIFFENERS - COMPRESSION FLANGE WEB YIELDING

$P_{bf} = \frac{4}{3}$ ((33.46 / 0.75) = 59.48K

Column Capacity = $F_y t_{wc} [t_{fb} + 6k + 2t_{ep} + 2w] = (36)(0.29)[0.38 + 6(1.063) + 2(\frac{3}{4}) + 2(\frac{1}{4})] = 91.4\text{K}$

P_{bf} < Column Capacity ∴STIFFENER NOT REQUIRED

8) COLUMN STIFFENERS - COMPRESSION FLANGE WEB BUCKLING

$$P_{bf} = \tfrac{4}{3}(33.46/0.75) = 59.48 \le \frac{4100(t_{wc})^3\sqrt{F_Y}}{\underset{\uparrow}{T}} = \frac{4100(0.29)^3\sqrt{36}}{7.625} = 78.68\text{K}$$

"T" Dimension in AISC W Shapes ∴STIFFENER NOT REQUIRED

9) COLUMN STIFFENERS - TENSION FLANGE LOCAL BENDING

P_{bf} = 4/3 (35.0 / 0.75) = 62.22K

$$0.4\sqrt{\frac{P_{bf}}{F_Y}} = 0.4\sqrt{\frac{62.22}{36}} = 0.525'' > t_{fc} = 0.435''$$ STIFFENERS ARE REQUIRED

Note: This need not be checked if stiffeners are provided for general flange bending.

10) COLUMN STIFFENERS - TENSION FLANGE OVERALL COLUMN FLANGE BENDING

This is per AISC (9th Edition pp 4-117)

$b_p = 2.5\,(\underbrace{P_f + t_{fb} + P_f}_{g_v}) = 2.5\,(3.5) = 8.75''$ (Effective Column Flange Length)

$C_a = 1.13$

$P_{eh} = (g_h/2) - k1 - d_{bo}/4 = (5.5/2) - 0.688 - (\frac{3}{4})/4 = 1.875''$

$C_b = 1.0$

$A_f / A_w = 1.0$

$\alpha_m = C_a C_b (A_f/A_w)^{1/3} (P_{eh}/d_{bo})^{1/4} = (1.13)(1.0)(1.0)(1.875/0.75)^{1/4} = 1.42$

$M_{eh} = \alpha_m F_t P_e / 4 = (1.42)(35.0)(1.875) / 4 = 23.3^{\text{"-K}}$

SAMPLE DESIGN 1 : END PLATE MOMENT CONNECTION–STANDARD FOUR TENSION BOLT

Column flange thickness required = $\sqrt{\dfrac{6M_{eh}}{27b_p}} = \sqrt{\dfrac{6(23.3)}{27(8.75)}}$ = 0.77" > 0.435"

STIFFENERS ARE REQUIRED.

11) <u>CHECK CAPACITY OF STIFFENED FLANGE</u>

The above AISC procedure calls for requirement of stiffeners. Capacity of stiffened flange is computed from formulas in the ASCE paper "Limit Design of Extended End Plate Connections" by Mann and Morris, ASCE ST3 March 1979.

$A = g_h = 5\frac{1}{2}$" B = End Plate width, b_{ep} = 8.0" $C = g_v$ = 3.5"

D = Bolt Diameter, $d_{bo} = \frac{3}{4}$" D' = Hole diameter = 0.8125"

n = CenterLine Bolt to edge of End Plate = (B - A) / 2 = (8.0 - 5.5) / 2 = 1.25"

m = (g_h / 2 - Col k_1) = (5.5/2 - 0.688) = 2.062"

n' = CenterLine bolt to edge of Col. Flange = (7.96 - 5.5) / 2 = 1.23"

V = P_f = Distance from top of stiffener / beam to centerline of bolt hole = 1.5"

$W = [m(m + n' - 0.5D')]^{1/2} = [2.062(2.062 + 1.23 - 0.5(0.8125))]^{1/2}$ = 2.44

F_{MS} = Limit State Capacity of a Stiffened Flange (i.e. without safety factor)

$$= \left[\left(\frac{1}{V} + \frac{1}{W}\right)(2m + 2n' - D') + \frac{2V + 2w - D'}{m} \right](t_{fc})^2 F$$

$$= \left[\left(\frac{1}{1.5} + \frac{1}{2.44}\right)(2(2.062) + 2(1.23) - 0.8125) + \frac{2(1.5) + 2(2.44) - 0.8125}{2.062} \right](0.435)^2(36)$$

= 65.67K (Yield)

Allowable flange capacity = $0.75F_{MS}$ = 0.75(65.67) = 49.25K

Actual Tension Flange Force = 35.0K Stiffened flange is O. K.

Additionally, check if the stiffened flange meets the requirement from the European code - ECCS.

The formula is modified for allowable stress and is similar to formula K1-1 for unstiffened flange.

Minimum Column Flange Thickness Required, $t_{fc} = 0.3\sqrt{\dfrac{P_{bf}}{F_Y}} = 0.3\sqrt{\dfrac{4/3(35/0.75)}{36}}$ = 0.394"

Actual flange thickness is 0.435" which exceeds 0.394"

∴Stiffened connection is O. K.

SAMPLE DESIGN 2 : END PLATE MOMENT CONNECTION–HAUNCHED FOUR TENSION BOLT

DESIGN DATA: (Refer To Figure 2)

Design the connection of sample design #1 without column stiffeners.

To accomplish this, a haunched connection is necessary. In addition, due to excessive column flange bending, the horizontal bolt gage (g_h) is reduced to the minimum $3\frac{1}{2}$".

Haunch Depth $D = 1.5d_b = 1.5(12.22) = 18.33"$

1) BEAM FLANGE FORCE

Modified Moment $M' = M + P^*(e)$ $\qquad e = 0.25d_b$

$$= 33.75 + 1.5(12.22 / (12)(4)) = 33.75 + 0.38 = 34.13^{'-K}$$

$$\text{Flange Force } F = \frac{M'}{(D - t_{fb})} \pm \frac{P}{2}$$

$$\text{Tension Flange } F_t = \frac{(34.13)(12)}{18.33 - 0.38} + \frac{1.5}{2} = 22.82 + 0.75 = 23.57K$$

Compression Flange $F_c = 22.82 - 0.75 = 22.07K$

2) BOLT STRESSES

$$\text{Bolt Shear Stress} = \frac{9.6}{(8)(0.4418)} = 2.72 \text{ KSI} < 21.0 \text{ KSI}$$ Table J3.2

$$\text{Allowable Tensile Stress} = \sqrt{44^2 - 4.39(f_v)^2} = \sqrt{[(44)(\tfrac{4}{3})]^2 - 4.39(\tfrac{2.72}{0.75})^2}\,(0.75) = 43.63 \text{ KSI}$$

(Table J3.3)

$$\text{Bolt Tensile Stress} = \frac{23.57}{4(0.4418)} = 13.34 \text{ KSI} < 43.63 \text{ KSI}$$

3) END PLATE TO BEAM FLANGE FILLET WELD

$$D_{W(REQD)} = \frac{F_t}{[2(b_{fb} + t_{fb}) - t_{wb}]0.928} = \frac{23.57}{[2(6.49 + 0.38) - 0.23]0.928} = 1.88 \text{ Sixteenths}$$

Minimum for $\frac{3}{4}$" thick end plate is $\frac{1}{4}$" fillet weld. (AISC J2.4)

4) END PLATE TO BEAM FILLET WELD

$$D_{W(REQD)} = \frac{0.6F_Y t_{wb}}{2(0.928)} = \frac{0.6(36)(0.23)}{2(0.928)} = 2.68 \text{ Sixteenths}$$

Minimum for $\frac{3}{4}$" thick end plate is $\frac{1}{4}$" fillet weld.

SAMPLE DESIGN 2 : END PLATE MOMENT CONNECTION–HAUNCHED FOUR TENSION BOLT

5) END PLATE WIDTH AND THICKNESS

Maximum Effective End Plate Width $b_{ep} = b_{fp} + 1 = 6.49 + 1 = 7.49"$

Use $7\frac{1}{2}"$ end plate width

For this, bolt edge distance = (7.5 - 3.5) / 2 = 2.0" OK

$P_e = P_f - d_{bo}/4 - 0.707w = 1.5 - (\frac{3}{4})/4 - 0.707(1/4) = 1.14$

$C_a = 1.13$

$C_b = \sqrt{b_{fb} / b_{ep}} = \sqrt{6.49 / 7.5} = 0.93$

A_f = Area of Tension Flange = $b_{fb}\, t_{fb} = (6.49)(0.38) = 2.47\ in^2$

A_W = Web Area Clear of Flanges = $(D - 2t_{fb})\, t_{wb} = (18.33 - 2(0.38))(0.23) = 4.04$ sq.in

$\alpha_m = C_a\, C_b\, (A_f / A_w)^{1/3}\, (P_e / d_{bo})^{1/4} = (1.13)(0.93)\,(2.47 / 4.4)^{1/3}\,(1.14 / 0.75)^{1/4} = 0.99$

$M_e = \alpha_m\, F_f\, P_e / 4 = (0.99)(23.57)(1.14) / 4 = 6.65^{"-K}$

End Plate Thickness Required = $\sqrt{\dfrac{6M_e}{b_{ep} 0.75 F_Y}} = 0.44"$

Use Minimum Thickness same as bolt diameter - 3/4" thick Plate

6) COLUMN WEB SHEAR

Column Web Shear = $\dfrac{F}{d_c t_{cw}} = \dfrac{23.57}{(9.73)(0.29)} = 8.35$ KSI < 14.4 KSI

7) COLUMN STIFFENERS - COMPRESSION FLANGE WEB YIELDING

$P_{bf} = \frac{4}{3}(22.07 / 0.75) = 39.24K$ Note: $F_c = 22.07K$

Column Capacity = $F_Y t_{wc}\,(t_{fb} + 6k + 2t_{ep} + 2w) = (36)(0.29)(0.38 + 6(1.063) + 2(\frac{3}{4}) + 2(\frac{1}{4})) = 91.4K$

P_{bf} < Column Capacity. STIFFENER NOT REQUIRED

8) COLUMN STIFFENERS - COMPRESSION FLANGE WEB BUCKLING

$P_{bf} = \frac{4}{3}(22.07 / 0.75) = 39.24K$ Note: $F_c = 22.07K$

Column Capacity = $\dfrac{4100(t_{wc})^3\sqrt{36}}{T} = \dfrac{4100(0.29)^3\sqrt{36}}{7.625} = 78.68K$

SAMPLE DESIGN 2 : END PLATE MOMENT CONNECTION--HAUNCHED FOUR TENSION BOLT

P_{bf} < Column Capacity. STIFFENER NOT REQUIRED

9) COLUMN STIFFENERS - TENSION FLANGE LOCAL BENDING

$P_{bf} = \frac{4}{3}(23.57 / 0.75) = 41.90K$ Note: $F_t = 23.57K$

Flange Thickness REQD $= 0.4\sqrt{P_{bf} / F_Y} = 0.4\sqrt{41.9 / 36.0} = 0.432" < t_{fc} = 0.435"$

STIFFENER NOT REQUIRED

10) COLUMN STIFFENERS - TENSION FLANGE OVERALL COLUMN FLANGE BENDING

This is per AISC (9th Edition pp 4-117)

$$b_p = 2.5\,(\underbrace{P_f + t_{fb} + P_f}_{g_v}) = 2.5\,(3.5) = 8.75"$$

$C_a = 1.13$

$C_b = 1.0$

$P_{eh} = g_h/2 - k_1 - d_{bo}/4 = (3.5)/2 - 0.688 - (\frac{3}{4})/4 = 0.8745"$

$A_f / A_w = 1.0$

$\alpha_m = C_a C_b\,(A_f / A_w)^{1/3}\,(P_e / d_{bo})^{1/4} = (1.13)(1)^{1/3}\,(0.8745 / 0.75)^{1/4} = 1.174$

$M_{eh} = \alpha_m F_t P_{eh} / 4 = (1.174)(23.57)(0.8745) / 4 = 6.05^{"\text{-}K}$

Required Column Flange Thickness $= \sqrt{\dfrac{6M_{eh}}{b_p 0.75 F_Y}} = \sqrt{\dfrac{6(6.05)}{(8.75)(27)}} = 0.392" < t_{fc} = 0.435"$

STIFFENERS NOT REQUIRED

NOTE: REFER TO THE STANDARD CONNECTION DETAIL AND CALL OUT ON ENGINEERING SKETCHES.

SAMPLE DESIGN 1 : END PLATE MOMENT CONNECTION–STIFFENED EIGHT TENSION BOLT

DESIGN DATA (Refer to Figure 3)

BEAM SIZE: W18 X 71

d_b = 18.47"

b_{fb} = 7.635"

t_{fb} = 0.810"

t_{wb} = 0.495"

COLUMN SIZE: W12 X 53

d_c = 12.060"

c_{fw} = 9.995"

t_{fc} = 0.575"

t_{wc} = 0.345"

k = 1.250"

k_1= 0.813

g_v = 4.0"

P_B = 3.0" (CenterLine outer row to CenterLine inner row of bolts)

g_h = 5.5" (Standard gage for W12 X 53 Column)

Bolts are A325N $\frac{3}{4}$" ∅

Total Number of Bolts = 12
(8 at top and 4 at bottom)

F_Y = 36 KSI

LOADS:

Beam Shear (V) = 25.0K

Beam Moment (M) = 150.0'-K

Beam Axial (P) = 0.0K

These forces have already been reduced.

This is equivalent to $\frac{1}{3}$ increase in allowable stresses

Note: For wind moment reversals, the tension at the bottom is much lower and four bolts suffice. Four bolts are insufficient at top. The bolt tensile stress was 57.7 KSI. Therefore 8 tension bolt connection is chosen.

Note: Refer to the standard connection detail and call out on engineering sketches.

1) BEAM FLANGE FORCE (Same as Standard 4 Tension Bolt Connection)

$$\text{Flange Force, } F = \frac{M}{d_b - t_{fb}} \pm \frac{P}{2}$$

Tension Flange F_t = (150)(12) / (18.47 - 0.810) = 101.93K

Compression Flange F_c = 101.93K

SAMPLE DESIGN 1 : END PLATE MOMENT CONNECTION–STIFFENED EIGHT TENSION BOLT

2) ESTIMATE BOLT FORCE AND CHOOSE TRIAL BOLT SIZE

Bolt Shear Stress $= \dfrac{25}{(12)(0.4418)} = 4.72$ KSI < 21 KSI

(8 Bolts top, 4 Bolts bottom)

Allowable Tensile Stress = $\sqrt{44^2 - 4.39(f_v)^2} = \sqrt{[44(\frac{4}{3})]^2 - 4.39(\frac{4.72}{0.75})^2}\ (0.75) = 42.87$ KSI

Table J3.3

Allowable Bolt Tension = (42.87)(0.4418) = 18.94K

(1st Trial Assume 6.8 Bolts Effective)

Bolt Force for Trial Bolt Size $= \dfrac{101.93}{6.8} = 14.99$ K < 18.94 K

Trial size of $\frac{3}{4}$"Ø may be adequate.

3) CALCULATE END PLATE THICKNESS REQUIRED FOR STIFFNESS

$P_f = 1.5$ $g_h = 5.5$

$d_{bo} = \frac{3}{4}$" $t_s = \frac{1}{2}$" (Stiffener)

$b_{ep} = 7.635 + 1" = 8.635$ (Maximum Effective End Plate Width)

$$t_{P1} = \frac{0.00885\,(P_f)^{0.873}\,(g_h)^{0.577}\,(F_t)^{0.917}}{(d_{bo})^{0.924}\,(t_s)^{0.112}\,(b_{ep})^{0.682}} = \frac{0.00885\,(1.5)^{0.873}\,(5.5)^{0.577}\,(101.93)^{0.917}}{(3/4)^{0.924}\,(1/2)^{0.112}\,(8.635)^{0.682}} =$$

$$= \frac{2.341}{3.085} = 0.759"$$

4) CALCULATE END PLATE THICKNESS REQUIRED FOR STRENGTH

$$t_{P2} = \frac{0.00625\,(P_f)^{0.257}\,(g_h)^{0.148}\,(F_T)^{1.017}}{(d_{bo})^{0.719}\,(t_s)^{0.162}\,(b_{ep})^{0.319}} = \frac{0.00625\,(1.5)^{0.257}\,(5.5)^{0.148}\,(101.93)^{1.017}}{(3/4)^{0.719}\,(1/2)^{0.162}\,(8.635)^{0.319}} =$$

$$= \frac{0.984}{1.446} = 0.681"$$

5) END PLATE THICKNESS

Thickness = Maximum of t_{P1} and t_{P2} = 0.759"

Minimum End Plate Thickness is same as bolt diameter.

Use 7/8" thickness End Plate

SAMPLE DESIGN 1 : END PLATE MOMENT CONNECTION–STIFFENED EIGHT TENSION BOLT

End Plate Width = 7.635 + 1 = 8.635

$\boxed{\text{Use 8 3/4"}}$

6) CALCULATE ULTIMATE BOLT FORCE

$$T_{ULT} = \frac{1.381\times10^{-4}\,(P_f)^{0.591}\,(F_T)^{2.583}}{(t_{ep})^{0.885}\,(d_{bo})^{1.909}\,(t_s\,)^{0.327}\,(b_{ep})^{0.965}} + P_T = \frac{1.381\times10^{-4}\,(1.5)^{0.591}\,(101.93)^{2.583}}{(7/8)^{0.885}\,(3/4)^{1.909}\,(1/2)^{0.327}\,(8.635)^{0.965}} + 28$$

$$= \frac{27.02}{3.275} + 28 = 8.25 + 28 = 36.25\text{ K}$$

↑Bolt Pretension AISC Table J3.7 9th Edition

7) CHECK BOLT SIZE

$T_{ULT} = 36.25 > 2 \times T_{ALLOW} = 2(18.94) = 37.88\text{K}$ (Step 2)

Prying action effects and bolt pretension are already accounted for in above equations. 8~3/4"Ø bolts for tension O. K.

8) END PLATE TO BEAM FLANGE FILLET WELD (Same as standard 4 tension bolt)

$$D_{W(REQD)} = \frac{F_t}{[2\,(b_{fb} + t_{fb}) - t_{wb}]0.928} = \frac{101.93}{[2\,(7.635 + 0.810) - 0.495]0.928} = 6.7 \text{ sixteenths}$$

$\boxed{\text{Use 7/16" Weld}}$

End Plate to Beam Web Fillet Weld (Same as standard 4 tension bolt)

$$D_{W(REQD)} = \frac{0.6F_Y\,t_{wb}}{2(0.928)} = \frac{0.6\,(36)(0.495)}{2(0.928)} = 5.76 \text{ Sixteenths}$$

$\boxed{\text{Use 3/8" Weld}}$

9) END PLATE TO BEAM FLANGE STIFFENER WELD

$\boxed{\text{Use 3/8" Fillet Weld}}$

10) COLUMN WEB SHEAR (Same as standard 4 tension bolt) Commentary E6

$$\text{Column Web Shear} = \frac{F}{d_c\,t_{cw}} = \frac{101.93}{(12.060)(0.345)} = 24.50\text{ KSI} > 0.4(36) = 14.4\text{ KSI}$$

Note: Commentary E6 can be used to reduce column web shear by taking advantage of story shear in column.

$$\text{Column Web Thickness REQD} = \frac{101.93}{(12.060)(14.4)} = 0.587"$$

Thickness of Doubler Plate REQD = 0.587 - 0.345 = 0.24

SAMPLE DESIGN 1 : END PLATE MOMENT CONNECTION–STIFFENED EIGHT TENSION BOLT

Use 1/4" Doubler Plate

11) COLUMN STIFFENERS - Compression Flange (Same as standard 4 tension bolt)

$P_{bf} = (\frac{4}{3})\ (101.93 / 0.75) = 181.2K$

For Web Yielding, Allowable = $F_Y\, t_{wc}\, (t_{fb} + 6k + 2t_p + 2w)$ (AISC pp4-117)

$= (36)(0.345)(0.810 + 6(1.25) + 2(\frac{7}{8}) + 2(\frac{7}{16})) = 135.8K$

For Web Buckling, Allowable = $\dfrac{4100(t_{wc})^3\sqrt{F_Y}}{T} = \dfrac{4100(0.345)^3\sqrt{36}}{9\frac{1}{2}(\leftarrow \text{AISC W Shapes})} = 106.3K$

(AISC EQ K1-8)

P_{bf} > Allowable. Therefore STIFFENERS ARE REQUIRED.

Area of Stiffeners REQD = (181.2 - 106.3) / 36 = 2.08 in^2

} Similar to K1-8

Thickness of Stiffeners REQD = 2.08 / (7.635 - 0.495) = 0.291"

Use 3/4" thick Stiffeners - Minimum is beam flange thickness rounded down to $\frac{1}{8}$

12) COLUMN STIFFENERS - TENSION FLANGE (Same as standard 4 tension bolt)

"Local Flange Bending" need not be checked for this connection since the load is spread out over longer length.

Overall flange bending needs to be checked (AISC 4-122)

$b_p = \underbrace{t_{fb} + 2P_f}_{g_v} + 3.5P_b = 4.0 + 3.5(3) = 14.5"$ (Effective Column Flange Length)

$C_a = 1.13$

$P_{eh} = (g_h / 2) - k_1 - d_{bo}/4 = (5.5 / 2) - 0.813 - (\frac{3}{4}) / 4 = 1.75"$

$C_b = 1.0$ $A_f / A_w = 1.0$

$\alpha_m = C_a\, C_b\, (A_f / A_w)^{1/3}\, (P_{eh} / d_{bo})^{1/4} = (1.13)(1.0)(1.0)^{1/3}(1.75 / 0.75)^{1/4} = 1.397$

$M_{eh} = \alpha_m\, F_f\, P_{eh} / 4 = (1.397)(101.93)(1.75) / 4 = 62.28^{\text{"-K}}$

Column Flange Thickness REQD = $\sqrt{\dfrac{6M_{eh}}{b_p 0.75F_Y}} = \sqrt{\dfrac{6(62.28)}{(14.5)(27)}} = 0.98 > t_{fc} = 0.575"$

SAMPLE DESIGN 1 : END PLATE MOMENT CONNECTION--STIFFENED EIGHT TENSION BOLT

PROVIDE STIFFENERS

USE 3/4" STIFFENERS

- At present, capacity of stiffened flange is not available. The column flange bending for this case is less severe than the 4 tension bolt connection.
- The minimum thickness with stiffeners is not applicable to this connection. (European code ECCS formula)
- AISC sponsored research and testing of this connection was used in the derivation of the above criteria.

NOTE: The need for column web reinforcement and stiffeners could have been eliminated using a "Haunched" Stiffened 8 Tension Bolt. The haunch preferred depth is $1.5d_b$. The haunched connection could use 8 bolts at top and 4 bolts at bottom.